Lecture Notes: RMS→RMS Norms, μP, and the Muon Optimizer

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Contents

1 Key concepts and terms (quick glossary)

- **Vector RMS norm.** For $x \in \mathbb{R}^d$,

$$\|x\|_{\mathrm{RMS}} \coloneqq \sqrt{\frac{1}{d}\sum_{i=1}^{d} x_i^2} \;=\; \frac{1}{\sqrt{d}}\,\|x\|_2\,.$$

- **Induced/operator norm.** For a matrix A and vector norms $\|\cdot\|_\alpha, \|\cdot\|_\beta$,

$$\|A\|_{\alpha\to\beta} \coloneqq \max_{x\neq 0} \frac{\|Ax\|_\beta}{\|x\|_\alpha} \;=\; \max_{\|x\|_\alpha=1} \|Ax\|_\beta\,.$$

- **Spectral norm.** $\|A\|_2 = \sigma_{\max}(A) = \max_{\|x\|_2=1} \|Ax\|_2$.

- **Frobenius norm.** $\|A\|_F = \sqrt{\sum_{ij} a_{ij}^2} = \sqrt{\sum_k \sigma_k(A)^2} = \sqrt{\mathrm{tr}(A^\top A)}$.

- **Semi-orthogonal matrix.** A rectangular matrix $Q \in \mathbb{R}^{m\times n}$ is *semi-orthogonal* if $QQ^\top = I_m$ (when $m \le n$) or $Q^\top Q = I_n$ (when $n \le m$). Such matrices have condition number 1 on their nonzero singular values.

- **Optimizer recipe (constrained linearization).** Choose a norm to constrain the update ΔW, then solve a local linear objective $\min_{\|\Delta W\|\le\eta} \langle \nabla_W \mathcal{L}(W),\, \Delta W\rangle$.

- **signSGD.** $W_{t+1} = W_t - \eta\,\mathrm{sgn}(g_t)$, where $g_t = \nabla_W \mathcal{L}(W_t)$ and $\mathrm{sgn}(\cdot)$ is applied elementwise.

- **Maximal Update Parameterization (μP).** A scaling/parameterization philosophy that aims to make activations and *updates* have stable ($\Theta(1)$) magnitudes as width changes, so hyperparameters transfer across model widths.

- **Shampoo.** A preconditioned optimizer using matrix inverse fractional powers of accumulated gradient covariance (left and right).

- **Muon.** *Momentum Orthogonalized by Newton–Schulz*: uses momentum plus an efficient approximate "semi-orthogonalization" (replacing singular values by ≈ 1) via Newton–Schulz polynomial iterations, avoiding explicit SVD.

2 Lecture 6: RMS→RMS norms and why they show up

2.1 Induced RMS→RMS operator norm

Let $A \in \mathbb{R}^{d_{\mathrm{out}}\times d_{\mathrm{in}}}$ map $x \in \mathbb{R}^{d_{\mathrm{in}}}$ to $Ax \in \mathbb{R}^{d_{\mathrm{out}}}$. Define the induced RMS→RMS norm:

$$\|A\|_{\mathrm{RMS}\to\mathrm{RMS}} \coloneqq \max_{\|x\|_{\mathrm{RMS}}=1} \|Ax\|_{\mathrm{RMS}}\,.$$

Because $\|x\|_{\mathrm{RMS}} = \|x\|_2/\sqrt{d_{\mathrm{in}}}$ and $\|Ax\|_{\mathrm{RMS}} = \|Ax\|_2/\sqrt{d_{\mathrm{out}}}$,

$$\begin{aligned}\|A\|_{\mathrm{RMS}\to\mathrm{RMS}} &= \max_{\|x\|_2=\sqrt{d_{\mathrm{in}}}} \frac{1}{\sqrt{d_{\mathrm{out}}}}\,\|Ax\|_2 \\ &= \sqrt{\frac{d_{\mathrm{in}}}{d_{\mathrm{out}}}} \max_{\|u\|_2=1} \|Au\|_2 \;=\; \sqrt{\frac{d_{\mathrm{in}}}{d_{\mathrm{out}}}}\;\|A\|_2\,.\end{aligned}$$

Takeaway: the induced RMS→RMS norm is a *dimension-rescaled spectral norm.*

2.2 Motivation (Xavier-style thinking)

A common motivation for RMS scaling is that in many random-initialization regimes, keeping RMS magnitudes stable across layers is easier to reason about than keeping raw ℓ_2 norms stable, because fan-in/fan-out appear explicitly through the $\sqrt{d_{\text{in}}/d_{\text{out}}}$ factor.

3 The optimizer recipe as a norm-constrained problem

3.1 General template

Given current parameters W, consider a first-order approximation:

$$\mathcal{L}(W + \Delta W) \approx \mathcal{L}(W) + \langle \nabla_W \mathcal{L}(W),\, \Delta W \rangle\,.$$

To choose an update, pick a step-size budget η and a norm $\|\cdot\|$, then solve

$$\Delta W^\star \in \arg\min_{\|\Delta W\| \le \eta} \langle \nabla_W \mathcal{L}(W),\, \Delta W \rangle\,.$$

Interpretation: "Choose a geometry (norm), then move as far as allowed in the steepest descent direction under that geometry."

3.2 Example: spectral-norm constrained update yields $UV^\top$

Let $G = \nabla_W \mathcal{L}(W) \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ with SVD $G = U\Sigma V^\top$. If we constrain $\|\Delta W\|_2 \le \eta$ (spectral norm), the optimizer of the linear objective is

$$\Delta W^\star = -\eta\, UV^\top,$$

a **semi-orthogonal update** (singular values all equal to 1 on the support of G).

4 Using RMS→RMS in the recipe gives layer-wise effective step sizes

Suppose we constrain the *RMS→RMS* operator norm of the update:

$$\|\Delta W\|_{\text{RMS}\to\text{RMS}} \le \gamma.$$

Using $\|\Delta W\|_{\text{RMS}\to\text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}}\,\|\Delta W\|_2$, this is equivalent to

$$\|\Delta W\|_2 \le \gamma \sqrt{\frac{d_{\text{out}}}{d_{\text{in}}}}.$$

Key observation: with a single global hyperparameter γ, different layers (different fan-in/out) effectively have different spectral-norm budgets, hence *layer-specific effective learning rates*. This is one route toward width-aware scaling rules.

5 Batch size 1, rank-1 gradients, and signSGD scaling

5.1 Why the per-layer gradient is rank 1 (batch size 1)

For a linear layer $h = Wx$ with loss $\ell(h)$, the gradient is

$$\nabla_W \ell = \frac{\partial \ell}{\partial h} x^\top,$$

an outer product of a d_{out} vector and a d_{in} vector, hence **rank 1**. This is the "low-rank gradient" intuition used in the notes.

5.2 Rank-1 fact: spectral norm equals Frobenius norm

If A has rank 1 with singular values $\{\sigma, 0, 0, \dots\}$, then

$$\|A\|_2 = \sigma = \|A\|_F .$$

5.3 Applying the RMS→RMS constraint to signSGD

Let $S = \operatorname{sgn}(G)$ be the elementwise sign of the gradient. In the batch-size-1 rank-1 setting, the notes argue S is also rank 1 (up to sign-pattern structure), so $\|S\|_2 = \|S\|_F$. Since entries of S are ± 1, we have

$$\|S\|_F^2 = \sum_{i=1}^{d_{\text{out}}} \sum_{j=1}^{d_{\text{in}}} 1 = d_{\text{out}} d_{\text{in}} \quad \Rightarrow \quad \|S\|_2 = \sqrt{d_{\text{out}} d_{\text{in}}}.$$

With signSGD update $\Delta W = \eta S$,

$$\|\Delta W\|_{\text{RMS}\to\text{RMS}} = \eta \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \, \|S\|_2 = \eta \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \sqrt{d_{\text{out}} d_{\text{in}}} = \eta \, d_{\text{in}}.$$

To keep $\|\Delta W\|_{\text{RMS}\to\text{RMS}} \le \gamma$, it suffices to choose

$$\boxed{\eta \;\le\; \frac{\gamma}{d_{\text{in}}}}$$

so the learning rate scales like $1/d_{\text{in}}$.

Conservativeness beyond batch size 1. As batch size increases, gradients become higher-rank. In general $\|A\|_2 \le \|A\|_F$, so replacing the spectral norm by Frobenius norm yields an *upper bound* and can be conservative. The notes remark that improved approximations to semi-orthogonal updates can reduce this conservativeness.

6 Maximal Update Parameterization (μP): intuition and desiderata

6.1 A scalar warm-up: choosing the "right units"

Consider i.i.d. $z_1, \dots, z_n$ with mean 0 and variance 1. Then

$$\frac{1}{\sqrt{n}} \sum_{i=1}^{n} z_i \Rightarrow \mathcal{N}(0, 1),$$

while $\frac{1}{n}\sum z_i \to 0$ and $\sum z_i$ blows up in scale. If an objective depends on a scaled sum, choosing the scaling factor (here $1/\sqrt{n}$) can make the large-n behavior nontrivial and stable.

6.2 Reparameterization idea

If you have an objective $F_n(c) = \mathbb{E}\big[f(c(z_1 + \cdots + z_n))\big]$, a reparameterization $c = \alpha/\sqrt{n}$ can yield a stable limit:

$$G_n(\alpha) \coloneqq F_n(\alpha/\sqrt{n}) \to \mathbb{E}\big[f(\alpha \cdot \mathcal{N}(0,1))\big],$$

so the optimal $\alpha^\star$ transfers across n (at least approximately), whereas the optimal $c_n^\star$ may not.

6.3 Feature-learning desiderata for a linear layer

For layer ℓ (ignoring bias and nonlinearity), $h_\ell = W_\ell h_{\ell-1}$ with $W_\ell \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$. Two desiderata emphasized in the notes are:

$$\text{(D1)} \quad \|h_\ell\|_{\text{RMS}} = \Theta(1), \qquad \text{(D2)} \quad \|\Delta h_\ell\|_{\text{RMS}} = \Theta(1) \text{ per step.}$$

A convenient sufficient condition is to keep both the weights and their updates controlled in RMS→RMS norm:

$$\boxed{\|W_\ell\|_{\text{RMS}\to\text{RMS}} = \Theta(1), \qquad \|\Delta W_\ell\|_{\text{RMS}\to\text{RMS}} = \Theta(1).}$$

Sketch of why: using $\Delta h_\ell = \Delta W_\ell \, h_{\ell-1} + W_\ell \, \Delta h_{\ell-1}$ and submultiplicativity,

$$\|\Delta h_\ell\|_2 \le \|\Delta W_\ell\|_2 \, \|h_{\ell-1}\|_2 + \|W_\ell\|_2 \, \|\Delta h_{\ell-1}\|_2 \,,$$

and converting between ℓ_2 and RMS introduces explicit fan-in/out scaling; controlling RMS→RMS norms keeps the recursion stable across widths.

7 Lecture 7: from Shampoo to Muon

7.1 Shampoo (matrix preconditioning)

Let $G_t = \nabla_W \mathcal{L}(W_t)$. Shampoo maintains left/right accumulators

$$L_t = L_{t-1} + G_t G_t^\top, \qquad R_t = R_{t-1} + G_t^\top G_t,$$

and applies the update

$$W_{t+1} = W_t - \eta \, L_t^{-1/4} \, G_t \, R_t^{-1/4}.$$

Intuition: the preconditioner reduces anisotropy so steps are more "uniform" across directions.

7.2 A revealing special case: no accumulation

If we ignore accumulation and use $L = GG^\top$ and $R = G^\top G$, then

$$\Delta W \propto (GG^\top)^{-1/4} \, G \, (G^\top G)^{-1/4}.$$

With SVD $G = U\Sigma V^\top$ (assume full rank for simplicity),

$$GG^\top = U\Sigma^2 U^\top, \qquad G^\top G = V\Sigma^2 V^\top,$$

so

$$(GG^\top)^{-1/4} G (G^\top G)^{-1/4} = U\Sigma^{-1/2}U^\top \cdot U\Sigma V^\top \cdot V\Sigma^{-1/2}V^\top = UV^\top.$$

So "Shampoo without accumulation" produces the same **semi-orthogonal direction** $UV^\top$ that arises from a spectral-norm constrained step.

7.3 Why compute $UV^\top$?

Replacing Σ by the identity removes domination by the largest singular value, leading to condition number 1 (on supported subspace) and a more uniform step across directions.

8 Muon: Momentum Orthogonalized by Newton–Schulz

8.1 High-level algorithm

Muon maintains a momentum-like buffer and then approximately "orthogonalizes" it.

Algorithm 1 Muon (conceptual form)

Require: learning rate η, momentum $\mu \in [0, 1)$, iteration count K

initialize $B_0 \leftarrow 0$
for $t = 1, 2, \dots$ **do**
  $G_t \leftarrow \nabla_W \mathcal{L}(W_t)$
  $B_t \leftarrow \mu B_{t-1} + G_t$
  $O_t \leftarrow \textsc{NewtonSchulzOrthogonalize}(B_t; K)$ ▷ $O_t \approx UV^\top$
  $W_{t+1} \leftarrow W_t - \eta\, O_t$
end for

8.2 Newton–Schulz as "polynomial on singular values"

A key trick: odd polynomials of the form

$$p(X) = a_0 X + a_1 X(X^\top X) + a_2 X(X^\top X)^2 + \cdots$$

commute with the SVD. If $X = U\Sigma V^\top$, then

$$p(X) = U\, p(\Sigma)\, V^\top,$$

so applying p changes singular values but preserves singular vectors.

A classic Newton–Schulz polynomial for pushing singular values toward 1 (for $\sigma \in (0, 1)$) is

$$p(\sigma) = \frac{3}{2}\sigma - \frac{1}{2}\sigma^3,$$

leading to the matrix iteration

$$X_{k+1} = \frac{3}{2}X_k - \frac{1}{2}X_k X_k^\top X_k.$$

Iterating drives Σ_k toward I (approximately), hence $X_k \approx UV^\top$.

8.3 Why normalization is needed

The iteration is stable when singular values lie in a suitable range (heuristically $(0, 1)$ in the notes). A simple way to enforce this is

$$X_0 \leftarrow \frac{B}{\|B\|_F},$$

because $\|B\|_2 \le \|B\|_F$, so all singular values of X_0 are ≤ 1.

8.4 Practical polynomials (NanoGPT example)

The notes mention using a higher-order odd polynomial that better matches the "spectrum" encountered in practice (e.g. NanoGPT):

$$f(\sigma) = 3.4442\,\sigma - 4.7750\,\sigma^3 + 2.0315\,\sigma^5.$$

This may sacrifice strict convergence guarantees for speed/quality of approximation; the goal is a good-enough semi-orthogonal direction.